

# DNA replication origin activation in space and time

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**Abstract** | DNA replication begins with the assembly of pre-replication complexes (pre-RCs) at thousands of DNA replication origins during the G1 phase of the cell cycle. At the G1–S-phase transition, pre-RCs are converted into pre-initiation complexes, in which the replicative helicase is activated, leading to DNA unwinding and initiation of DNA synthesis. However, only a subset of origins are activated during any S phase. Recent insights into the mechanisms underlying this choice reveal how flexibility in origin usage and temporal activation are linked to chromosome structure and organization, cell growth and differentiation, and replication stress.

## Replication forks

Structures that are created by unwinding of the double helix at a replication origin, from which DNA synthesis will progress in opposite directions.

DNA replication in eukaryotic cells requires the accurate synthesis of large amounts of DNA. Moreover, to produce an adult organism from a single fertilized oocyte, DNA must be replicated many times to give rise to the  $4 \times 10^{13}$  copies of DNA that make up a human body. Errors in DNA replication can be amplified and accumulate over time, leading to genome instability, which has deleterious consequences for organs and tissues and is a hallmark of cancer<sup>1</sup>. DNA replication errors also accumulate in stem cells of adult organisms and are associated with ageing<sup>2</sup>. Furthermore, although bacterial DNA replication is coupled only to cell growth, metazoan DNA replication is coupled to both cell growth and cell differentiation. Thus, chromatin features and chromosome organization that determine cell identity also have to be tightly regulated and accurately reproduced during development.

Eukaryotic cells have developed several mechanisms to preserve genome stability, which include ensuring fidelity of DNA replication at replication forks. The control of activation of DNA replication origins is now recognized as an important mechanism that is used by eukaryotic cells to adapt to their environment, to tissue-specific transcriptional programmes and to constraints that are linked to the complex structures and variety of conformations of chromosomes.

DNA replication origins are usually defined as the genomic regions at which DNA replication starts. However, they encompass at least two distinct elements: the DNA region that is recognized and bound by specific proteins and that will form the pre-replication complex (pre-RC) (FIG. 1) and the downstream site of initiation of DNA synthesis to which the DNA polymerase machinery is recruited. In *Saccharomyces cerevisiae*, replication

origins are identified by specific DNA sequences<sup>3</sup>, but the choice of origins that are activated is flexible (see below). Conversely, no universal signature or set of signatures that could predict all replication origins in a metazoan genome has yet been identified, although genome-wide analyses have shown that metazoan replication origins can have some preferred sequence motifs<sup>4</sup>.

Replication origins in eukaryotes are determined in two subsequent steps (FIG. 1): the recognition of the pre-RC site, a reaction known as replication origin ‘licensing’, and the activation of DNA synthesis, which is known as origin ‘firing’. This two-step mechanism is crucial for preventing re-replication within the same cell cycle (BOX 1).

Importantly, only a subset of all licensed origins is activated in a cell in any one cell cycle. The choice of origins to be activated varies from cell to cell, even in the same cell population, implying that origin usage is flexible in mammalian cells<sup>5</sup>, as is the case in yeast cells<sup>6,7</sup>.

This uncoupling between the first step of replication origin licensing and the second step of origin activation, and the flexibility in the usage of replication origins, are prominent and crucial features of eukaryotic DNA replication origins (for a review, see REF. 8). The uncoupling is essential at five levels: the adaptation of DNA replication to the structural organization of chromosomes; the specific temporal activation of origins during the cell cycle; the response of DNA replication to cell growth conditions; the response to DNA damage and the adaptation of DNA replication to gene expression and cell identity.

In this Review, we discuss how origin activation enables cells to adapt to changing chromatin and cellular environments. We first describe the biochemistry of

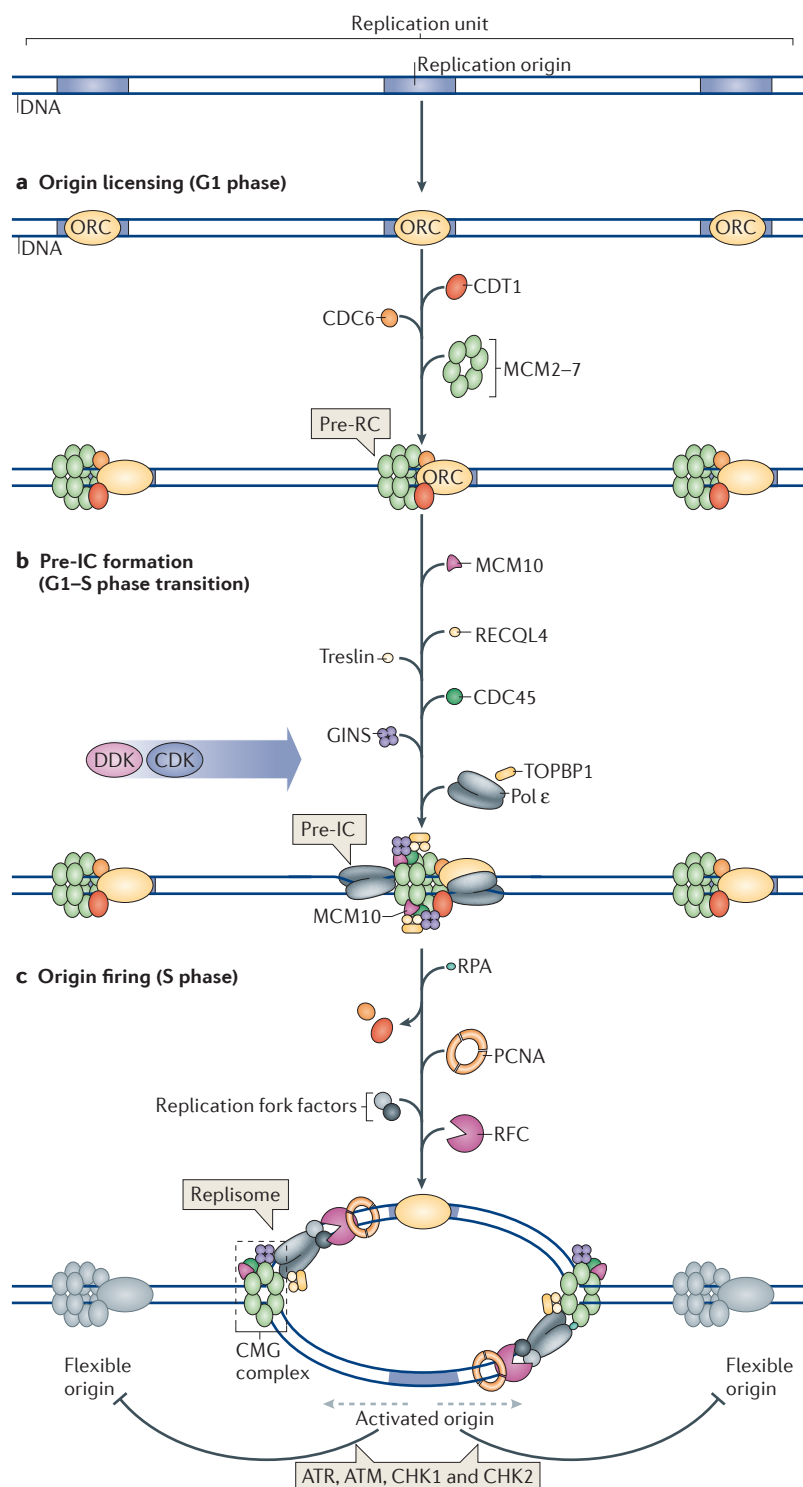
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origin activation, the genetic and epigenetic marks that may be associated with active origins, and how transcription might be involved in initiation of DNA replication. We then discuss how replication origin activation may be connected with the nuclear and chromosome structure. Finally, we focus on the links between origin activation, the control of its timing during the S phase and the replication checkpoint controls.

## DNA replication initiation: a two-step process

Although the origin licensing reaction takes place at all potential replication origins in the genome during the G1 phase of the cell cycle, only a subset of origins are activated at any time. Origin licensing and activation reactions have distinct biochemical features. The complete sequence of events has now been mimicked *in vitro*<sup>9</sup>.



**Figure 1 | Formation and activation of DNA replication origins.** The figure shows a replication unit with three potential replication origins. **a** | Licensing of replication origins is restricted to the G1 phase of the cell cycle and results from the sequential loading of pre-replication complex (pre-RC) proteins on all potential origins in the genome. First, the origin recognition complex (ORC), comprising the six subunits ORC1–6, which has ATPase activity, is recruited to replication origins. This is followed by the binding of CDC6 and CDC10-dependent transcript 1 (also known as DNA replication factor CDT1). Loading of the mini-chromosome maintenance (MCM) helicase complex, which contains the six subunits MCM2–7, is the last step of the licensing reaction and can take place only if ORC, CDC6 and CDT1 are already bound to origins. **b, c** | Origin activation involves the formation of a pre-initiation complex (pre-IC) and activation of the MCM helicase complex. Assembly of the pre-IC is triggered by DBF4-dependent kinase (DDK) and cyclin-dependent kinases (CDKs) at the G1–S phase transition, and its activation into a functional replisome occurs in the S phase. DDK and CDKs phosphorylate several replication factors (of which MCM10, CDC45, ATP-dependent DNA helicase Q4 (RECQL4), Treslin, GINS, DNA topoisomerase 2-binding protein 1 (TOPBP1) and DNA polymerase ε (Pol ε) are the most important) to promote their loading on origins. Moreover, DDK and CDKs directly phosphorylate several residues within the MCM2–7 complex, resulting in helicase activation and DNA unwinding. During helicase activation, the MCM2–7 double hexamer divides into two hexamers that function at the two replication forks emanating from the replication origin. Helicase activation induces the recruitment of other proteins (such as replication factor C (RFC), proliferating cell nuclear antigen (PCNA), replication protein A (RPA) and other DNA polymerases) that convert the pre-IC into two functional replication forks that move in opposite directions from the activated origin, with the replisome (a protein complex) at each replication fork. It is not yet clear whether one or two ORCs remain on the duplicated origin after initiation of DNA synthesis. The functional helicase at the forks is the CMG complex (which consists of CDC45, the MCM hexamer and the GINS complex). In a replication unit, only one out of three origins on average is activated, whereas the other adjacent origins remain silent, although they have been licensed. Therefore, a replisome is only formed in the activated origin. In a given cell population, different origins can be used in individual cells; thus, a cell population contains a range of flexible origins. Inhibition of adjacent origins within a replication unit is controlled in part by the checkpoint kinases Ser/Thr protein kinase ATR and Ser protein kinase ATM that activate checkpoint kinase 1 (CHK1) and CHK2. However, the exact mechanisms that are responsible for the local inhibition of these flexible origins remain unclear. Similarly, it has not yet been determined how flexible origins are selected for activation or silencing.

# Box 1 | Preventing origin re-replication

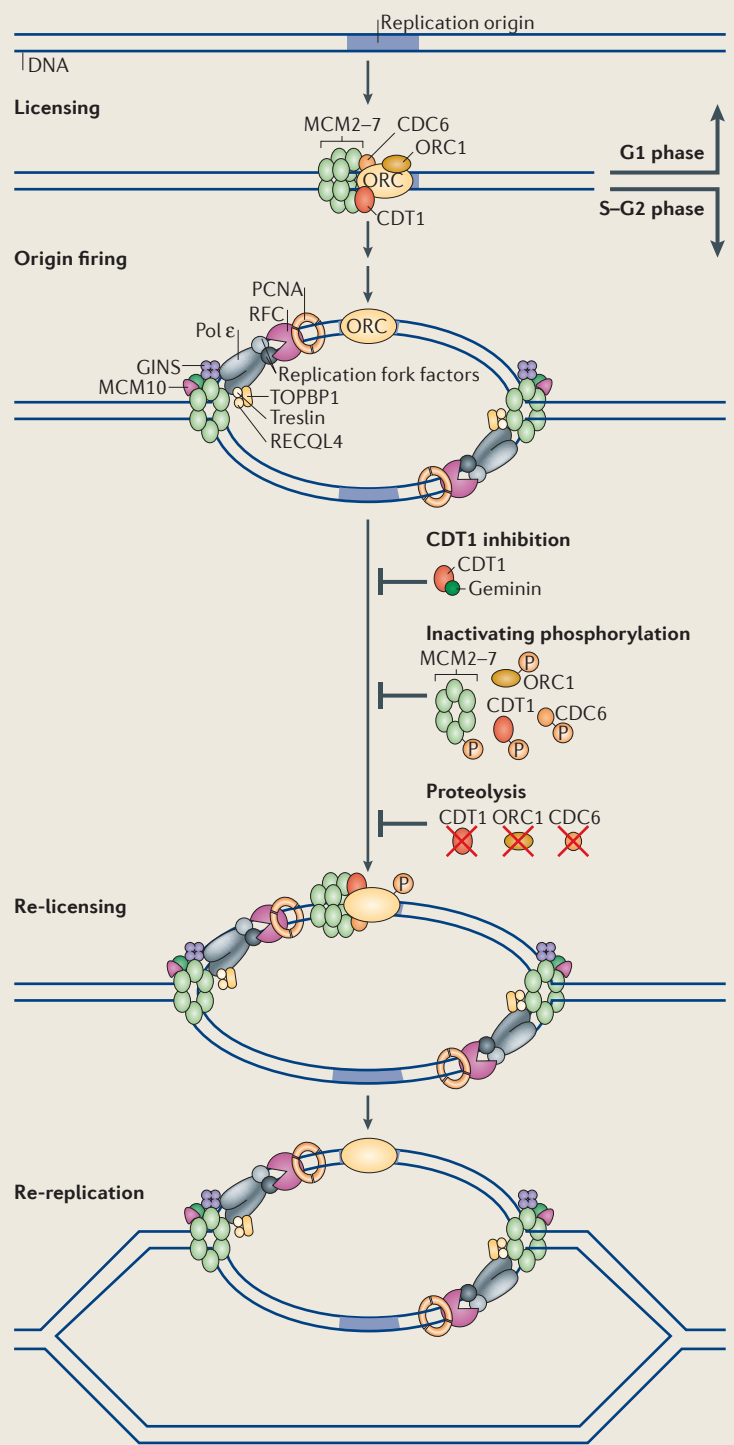
Replication origin licensing must occur only once per cell cycle (from late mitosis to late G1 phase) to ensure accurate duplication of the genome (see REF. 17 for a review). Inappropriate licensing (that is, licensing that takes place after the beginning of DNA synthesis), which may be caused, for instance, by overexpression of CDC10-dependent transcript 1 (also known as DNA replication factor CDT1), can lead to re-activation of origins that have already been used during that S phase and subsequently to genomic amplification, a process known as re-replication. Re-replicating cells show signs of DNA damage and genomic stress or instability<sup>166,167</sup>, which are associated with cell cycle arrest, senescence and apoptosis<sup>166,168,169</sup>.

To prevent re-replication, metazoan cells have evolved several mechanisms to specifically regulate origins that have already been activated. The three main mechanisms are illustrated in the figure. First, the coiled-coil protein geminin is a key factor for restraining origin re-establishment once replication has been initiated. It acts by directly binding to CDT1 to inhibit its licensing activity<sup>170,171</sup>. Second, phosphorylation (mainly mediated by cyclin-dependent kinases) of pre-RC components — such as the mini-chromosome maintenance (MCM) complex, comprising subunits MCM2–7 (REF. 172), origin recognition complex subunit 1 (ORC1)<sup>173</sup>, CDC6 (REF. 174) and CDT1 (REFS 175,176) — is involved in repressing reactivation of origins. Third, the ubiquitin–proteasome pathway contributes to regulation of licensing once cells enter the S phase. Cullin-based E3 ubiquitin ligases interact with ORC1 and CDT1 (which are targeted by SCF<sup>Skp2</sup> (SKP1–Cullin 1–S phase kinase-associated protein 2))<sup>177,178</sup> and CRL4<sup>Cdt2</sup> (also known as Cullin ring ligase 4 complex associated with denticleless protein homologue–CDC10-dependent transcript 2; CRL4DTL–CDT2)<sup>179,180</sup> for degradation. In addition, the anaphase-promoting complex APC/C and its regulators have an important role in regulating establishment of origins, both positively (by targeting geminin<sup>181</sup>) and negatively (by targeting CDC6 (REF. 182) and CDT1 (REF. 183)). Finally, the checkpoint proteins Ser/Thr protein kinase ATR and checkpoint kinase 1 (CHK1) also repress re-replication in normal cells<sup>184</sup>.

Notably, physiological re-replication can occur during development as a means to amplify all or specific genomic loci to regulate tissue organization (for a review, see REF. 185). In *Drosophila melanogaster* follicle cells, for example, re-replication at chorion genes facilitates the production of large amounts of eggshell proteins. Re-replication requires pre-RC components and is associated with high levels of histone acetylation of the amplified loci<sup>186,187</sup>.

Finally, CDC6 (REF. 188) and CDT1 (REF. 189) are proto-oncogenes, which highlights the possible link between excessive licensing and development of cancer. Interestingly, tumour-derived cells are more sensitive to re-replication than normal cells, leading to their preferential death<sup>166,190</sup> and thus opening the way for potential novel anticancer therapeutics.

PCNA, proliferating cell nuclear antigen; Pol ε, DNA polymerase ε; RECQL4, ATP-dependent DNA helicase Q4; RFC, replication factor C; TOPBP1, DNA topoisomerase 2-binding protein 1.



**The origin licensing reaction.** Origin licensing involves sequential and interdependent anchoring of different proteins (for a review, see REF. 10) (FIG. 1; TABLE 1). First, the origin recognition complex (ORC), which consists of ORC subunit 1 (ORC1) to ORC6 and has ATPase activity, is recruited to replication origins. ORC assembly on chromatin is followed by the binding of another ATPase,

CDC6. Then, CDC10-dependent transcript 1 (also known as DNA replication factor CDT1) is loaded onto origins. CDC6 and CDT1 recruit the mini-chromosome maintenance (MCM) complex, which is a hexamer composed of the six subunits MCM2–7, resulting in the formation of a pre-RC (FIG. 1). The DNA helicase activity of the MCM complex is essential for the unwinding of DNA<sup>10</sup>.

Table 1 | **Factors involved in DNA replication initiation in eukaryotes**

| Replication factor      | Name                                                                                | Function                                                                                                 | Complex                      | Conservation                                                         |
|-------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------|
| ORC1–6                  | Origin recognition complex (subunits 1 to 6)                                        | Initiates pre-RC formation; has an ATPase activity                                                       | Pre-RC                       | Archaea, yeast, flies, fishes, amphibians, birds, mammals and plants |
| LRWD1                   | Leucine-rich repeat and WD40 repeat-containing protein 1 (also known as ORCA)       | Associates with ORC and stabilizes it on chromatin                                                       | Pre-RC                       | Fishes, amphibians, birds and mammals                                |
| CDC6                    | Cell division control protein 6 homologue                                           | Component of the pre-RC; AAA+ ATPase                                                                     | Pre-RC                       | Archaea, yeast, flies, fishes, amphibians, birds, mammals and plants |
| CDT1                    | CDC10 dependent transcript 1                                                        | Interacts with CDC6 and MCM2–7 complex                                                                   | Pre-RC                       | Yeast, flies, fishes, amphibians, birds, mammals and plants          |
| MCM2–7                  | Mini chromosome maintenance proteins 2 to 7                                         | Replicative helicase; AAA+ ATPase                                                                        | Pre-RC, pre-IC and replisome | Archaea, yeast, flies, fishes, amphibians, birds, mammals and plants |
| Geminin                 | Geminin                                                                             | Replication inhibitor; binds to CDT1 and inhibits the formation of new pre-RCs during S phase            | Pre-RC (inhibitor)           | Flies, fishes, amphibians, birds and mammals                         |
| DDK                     | DBF4-dependent kinase (also known as the DBF4–CDC7 complex)                         | Helicase activation; phosphorylates MCM2–7                                                               | Pre-IC                       | Yeast, flies, fishes, amphibians, birds, mammals and plants          |
| CDKs                    | Cyclin-dependent kinases (active forms are heterodimers of cyclin and CDK subunits) | Helicase activation; phosphorylates treslin                                                              | Pre-IC                       | Yeast, flies, fishes, amphibians, birds, mammals and plants          |
| TOPBP1 (Dpb11 in yeast) | Topoisomerase II-binding protein 1                                                  | Helicase activation and replisome assembly; associates with treslin                                      | Pre-IC, replisome            | Yeast, flies, fishes, amphibians, birds and mammals                  |
| GINS                    | 5-1-2-3 in Japanese                                                                 | Helicase activation and replisome assembly (component of CMG complex); heterotetrameric pre-IC component | Pre-IC, replisome            | Archaea, yeast, flies, fishes, amphibians, birds, mammals and plants |
| CDC45                   | Cell division control protein 45 homologue                                          | Helicase activation and replisome assembly (component of CMG complex)                                    | Pre-IC, replisome            | Yeast, flies, amphibians, mammals and plants                         |
| Treslin                 | TOPBP1-interacting, replication-stimulating protein                                 | Helicase activation and replisome assembly; associates with TOPBP1                                       | Pre-IC, replisome            | Fishes, amphibians, birds and mammals                                |

CMG, CDC45–MCM–GINS complex; pre-IC, pre-initiation complex; pre-RC, pre-replication complex. Human nomenclature is used for proteins.

In yeast and mouse cells, CDT1 and the MCM hexamer form a complex before being loaded onto DNA at replication origins<sup>11–13</sup>. However, such stable complexes have not been observed in *Xenopus laevis* egg extracts that are competent for DNA replication<sup>14,15</sup> or in human cell extracts (S. Hills and J. F. Diffley, personal communication, 2015).

After replication origins are licensed, cells should prevent re-licensing during the S phase to ensure that chromosomes replicate only once per cell cycle<sup>16</sup>. This is achieved through several mechanisms, including interaction of CDT1 with its inhibitor geminin and CDT1 ubiquitylation and degradation during the S phase, as well as phosphorylation of several initiation factors<sup>17</sup> (BOX 1).

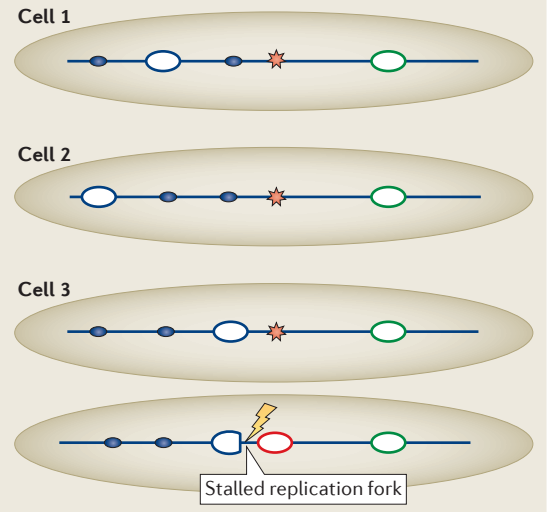
**Flexibility in the activation of replication origins.** Early studies of DNA replication reported that the number of replication origins that are licensed in the G1 phase in a given cell is greater than the number of origins that are activated during the S phase<sup>18,19</sup>. This led to the notion of replication origin efficiency to describe

the frequency at which a chosen origin is activated in a given cell in a given cell cycle. The different usage of replication origins permits their classification into three categories: constitutive, flexible and dormant (BOX 2). Therefore, only some of all potential origins are activated in proliferating cultured cells, and this proportion can vary from cell to cell within the same culture. In rapidly dividing cells of early *Drosophila melanogaster* and amphibian embryos, origin efficiency is extremely high<sup>20,21</sup>. The number of activated origins progressively decreases over time — in adult somatic cells only 20–30% of all potential replication origins are activated — which contributes to lengthening of the S phase.

Currently, a major challenge is to understand how origins that are to be activated in the S phase are chosen from all the potential origins that are assembled as pre-RCs in the G1 phase<sup>8</sup>. Is this choice entirely stochastic? Or is it dictated by features or constraints that are linked to nuclear organization and nuclear metabolism (including transcriptional activity and other chromatin-associated events) and/or by specific activating factors?

**Box 2 | Different types of replication origin according to their usage**

Replication origins fall into three categories (see the figure): constitutive origins (green circles), flexible origins (blue circles) and dormant origins (red stars). Constitutive origins are activated in all cells of a cell population and at all times (that is, independently of developmental stage and/or environmental conditions). These origins are only a small subset of all origins. Most of the origins in any cell population or organism are flexible origins: their use varies from cell to cell in an apparently stochastic manner. Thus, although only a few of them will fire in a given cell, all flexible origins in a cell population will be scored. Dormant origins are defined as DNA replication origins that are licensed but do not fire during a normal cell cycle<sup>191</sup>. These origins are activated following DNA damage that blocks a neighbouring replication fork (as shown in the lower cell). Filled circles indicate licensed origins that are not activated; open circles indicate active origins.



These questions are crucial because the large excess of licensed origins is an important feature of eukaryotic cells that enables them to respond to replication stress and to cell cycle checkpoints to prevent genome instability (see below).

**Biochemistry of the activation of replication origins.** Unlike pre-RC assembly at origins, which requires little or no cyclin-dependent kinase (CDK) activity, origin activation requires high levels of CDK activity. DNA replication origin activation is achieved through a highly regulated mechanism that involves a series of phosphorylation events on the subunits of the MCM helicase complex, mainly by DBF4-dependent kinase (DDK; also known as the CDC7–DBF4 complex) and by CDKs, which are Ser/Thr protein kinases. This regulated reaction involves ATP and several ATPase motifs on different MCM subunits<sup>22</sup>.

When the MCM complex is first loaded to form the pre-RC, it is an inactive head-to-head double hexamer that surrounds double-stranded DNA (FIG. 1). Activation of replication origins involves the dissociation of the double MCM hexamer into two active MCM hexamers that form the two replisomes that can unwind DNA and start two replication forks at each replication origin (FIG. 1). This step is induced by the transient binding of DDK to chromatin, which enables it to phosphorylate the MCM complex<sup>23–25</sup>. This phosphorylation is an essential step for the subsequent recruitment and formation of the CMG complex, which consists of CDC45, the MCM complex and the DNA replication complex GINS and participates in the formation of the pre-initiation complex (pre-IC), which is defined as the protein complex preceding the activation of the DNA helicase<sup>23,24,26–28</sup> (FIG. 1). However, phosphorylation by DKK alone is not sufficient to dissociate the MCM double hexamer that is loaded at replication origins<sup>29</sup>. CDKs are also needed for MCM phosphorylation and for binding of GINS and CDC45 to chromatin to establish the pre-IC. After being assembled and activated,

the CMG complex unwinds double-stranded DNA to initiate DNA synthesis. In vertebrates, CDKs phosphorylate treslin (which is orthologous to Sld3 in yeast) to promote its interaction with DNA topoisomerase 2-binding protein 1 (TOPBP1; orthologous to yeast Dbp11), a protein that is essential for the activation of the CMG helicase<sup>30–32</sup>. Treslin and TOPBP1 are components of the pre-IC that bind to the MCM hexamer. CDKs also phosphorylate and activate ATP-dependent DNA helicase Q4 (RECQL4; the vertebrate orthologue of yeast Sld2) that, with MCM10, is required for the formation of the CMG complex<sup>33,34</sup>.

**Genetic determinants affecting activation**

In budding yeast, DNA replication initiates at AT-rich autonomously replicating sequences (ARSs) that contain 11–17 bp consensus elements that are targeted by ORC. However, only a small proportion of these ARS consensus sequences are used as replication origins, suggesting that other elements contribute to origin recognition and activation. Recent studies have shown that different sequences can support plasmid DNA replication in yeast extracts, if pre-RCs are properly formed<sup>29,35</sup>.

In metazoans, replication origins tend to localize in CG-rich regions, such as CpG islands<sup>5,36–39</sup>. Experiments that showed that metazoan origins can initiate replication when transferred to ectopic sites provided strong evidence that genetic sequences may have an important role in origin identification<sup>40</sup>. However, strict consensus genetic determinants have been difficult to identify in metazoans. An origin G-rich repeated element (OGRE) was identified at most mammalian and *D. melanogaster* replication origins. OGREs can potentially form G quadruplexes (G4s)<sup>5,41</sup> that might affect origin positioning and efficiency<sup>42</sup>. Although replication origins contain this motif, potential G4s in the genome are in excess relative to activated origins. This suggests that other metazoan replication origin signatures are necessary, or that there is a large excess of potential origins that are never used in a normal cell cycle.

**CpG islands**

Short DNA sequences (at least 200 bp) with a high occurrence of CpG dinucleotides (for example, a ratio of CpG observed to CpG expected > 0.6). In vertebrates, these sequences are often present near the transcription initiation sites of housekeeping genes.

**G quadruplexes**

(G4s). Guanine-rich sequences that form a four-stranded DNA structure with tetrads of guanine stacking on top of each other. These have been well studied in telomere regions.



Interestingly, recent profiling of non-canonical fission yeast species such as *Schizosaccharomyces japonicus* and of the budding yeast *Pichia pastoris* also revealed the presence of GC-rich replication origins<sup>43</sup>, showing that the AT-richness of replication origins is not a universal feature among unicellular organisms. Moreover, it could be speculated that different signature combinations are required for the two-step process (consisting of origin licensing and origin activation) that leads to DNA replication. The synergic action of several factors could 'frame' the origins that are to be activated to distinguish them from other licensed origins and support their flexible selection in different chromatin contexts.

### Transcription and the chromatin environment

The regulation of DNA replication origins starts in the G1 phase of the cell cycle with the assembly of pre-RCs and extends through the whole S phase with the activation of replication origins. During the same period, DNA is also highly transcribed and chromatin modifications occur. Therefore, it is not surprising that some coordination and harmonization is detected between these nuclear events.

**Transcription and origin activation.** In eukaryotes, a relationship between transcription and replication origins was first described for the replication of small DNA viruses. The replication initiation factor is viral and is often a DNA helicase that has both transcription and replication functions or a DNA helicase associated with a viral transcription factor (reviewed in REF. 44). Early studies also showed that transcriptionally active genes are replicated early in the S phase, whereas transcriptionally inert genes replicate later<sup>45</sup>. In metazoans, genome-wide analyses have indicated that although replication origins are present throughout genes, those that are close to promoter regions are more concentrated and more active<sup>5,46,47</sup>.

Transcription may have two opposing effects on the selection of the origins to be activated. High levels of RNA synthesis could be an obstacle to the formation of a replication complex and could prevent pre-RC activation<sup>46,48</sup>. In this case, activation of an origin located upstream of the promoter could be preferentially activated. Conversely, the open chromatin configuration of transcriptionally active promoters may favour the selection of replication origins that are located in the same area. The frequent localization of replication origins at promoter regions might partly explain their preferential association with CpG islands, which are also associated with gene promoters<sup>49</sup>. However, CpG islands that are not located at promoters are also frequently associated with replication origins<sup>5</sup>.

Despite much effort, no clear interaction between basal transcription factors and replication initiation factors has been found. This does not exclude the possibility that such associations may occur on chromatin if transcription factor binding sites are close to replication origins or if chromatin looping brings them close to each other. This could have a cooperative effect on

the stabilization or activation of the replication initiation complex and could explain the increase in replication origin activity when a strong transcriptional promoter is set up on DNA<sup>50</sup>. Transcriptional regulators may positively<sup>51</sup> or negatively<sup>52</sup> affect the activation of specific origins. Although replication origins are preferentially active at promoter and gene-rich regions, they are infrequent at heterochromatin-rich regions, which are gene-poor and late-replicating regions<sup>5,53</sup>. However, it remains unclear whether DNA replication origins help to establish an active transcriptional promoter or the reverse, and this question may have different answers depending on the DNA region.

**Chromatin features affect origin activation.** From budding yeast to mammalian cells, active origins overlap with nucleosome-free regions, an environment that is similar to that found at promoters of transcriptionally active genes<sup>54–57</sup>. In yeast, nucleosome-free regions are associated with the ARS consensus sequence<sup>54,55,58</sup>, which determines the location of the origin of replication, and are flanked by positioned nucleosomes. In *D. melanogaster*, ORC is associated with open chromatin<sup>53</sup> at sites of histone hyperacetylation<sup>59</sup>. Conversely, in mammalian cells, the presence of a nucleosome is associated with replication initiation sites<sup>41,60</sup>. This apparent discrepancy could be explained if the nucleosome was associated only with the replication initiation site, whereas the upstream region, to which ORC binds, was nucleosome-free. This would mean that during most of the cell cycle, the pre-RC would be nucleosome-free, with a nucleosome positioned downstream that would be transiently released during initiation of DNA synthesis. This remains to be experimentally demonstrated.

Specific histone marks have been detected at origins. However, most of them are likely to be associated with pre-RC formation rather than origin activation, because activated origins are only a small proportion of all origins in a cell. The dynamics of histone modifications during the assembly and activation of replication origins is still unknown.

Histone acetylation, which is a mark for open chromatin, is commonly detected at replication origins. Histone acetyltransferase binding to ORC1 (HBO1; also known as histone acetyltransferase KAT7)<sup>61,62</sup> seems to be involved in origin activation, but it is still unclear whether its effect is mediated through histone or pre-RC acetylation<sup>63</sup>.

Histone 4 Lys 20 methylation (H4K20me) and its associated methylase *N*-Lys methyltransferase SETD8 (also known as PR-SET7) have also been detected at replication origins<sup>42,64,65</sup>, and the bromodomain homology domain of metazoan ORC1 seems to be involved in H4K20me2 recognition<sup>66</sup>. Methylation of histone H3 has also been associated with replication origins. These include H3K27me3, which is associated with origins that are activated in mid-S phase<sup>67</sup>; H3K56me1, which might be involved in recruiting proliferating cell nuclear antigen (PCNA), which is involved in replication activation, to chromatin<sup>68</sup>; and H3K79me2, which could help limiting re-replication events during the cell cycle<sup>69</sup>.

#### Open chromatin

A loose or partly decondensed chromatin structure, found in euchromatin regions that are permissive for transcription.

#### Proliferating cell nuclear antigen

(PCNA). A homotrimer that is involved in the processivity of DNA polymerases during DNA replication, as well as having a role in DNA repair.

In *D. melanogaster* and in plants, the histone variant H3.3, which is associated with active transcription, is enriched at origins<sup>53,70</sup>. Interestingly, both *D. melanogaster* and human ORC can interact with heterochromatin protein 1 homologue- $\alpha$  (HP1 $\alpha$ ; also known as chromobox protein homologue 5), which is a heterochromatin-specific protein and an important reader of the H3K9me2 and H3K9me3 epigenetic marks that are known to promote gene silencing<sup>71,72</sup>. Consistent with this observation, DNA replication origins on *D. melanogaster* chromosome 4, which is an almost completely heterochromatinized chromosome, show a strong association with HP1 sites but not with CpG-rich regions, as in the other chromosomes<sup>5</sup>. This suggests that metazoans are characterized by the presence of different classes of replication origin, the activation of which is regulated by different signatures.

### Non-coding RNAs and origin activation

A considerable part of metazoan genomes encodes non-coding RNAs (ncRNAs) that are involved in several biological processes. Recent studies have shown that ncRNAs can guide histone modifiers to specific chromatin regions or have structural functions in the spatial organization of chromatin (reviewed in REF. 73). ncRNAs also could have a role in origin recognition and/or activation. Specifically, in *Tetrahymena thermophila*, ORC contains an ncRNA that functions in origin recognition in ribosomal DNA (rDNA)<sup>74</sup>. Similarly, a G-rich RNA molecule is required to stabilize binding of ORC to a viral origin of replication<sup>75</sup>. This RNA forms a G4 structure that is essential for mediating the interaction between ORC and Epstein–Barr nuclear antigen 1, which is required for recruitment of ORC to the viral chromatin<sup>76</sup>. It has also been shown that ORC preferentially binds to G4 RNA rather than to single-stranded DNA<sup>77</sup>.

Moreover, a class of small ncRNAs, called Y RNAs, associate with chromatin in the G1 phase before initiation of DNA replication<sup>78</sup>. However, it is not yet clear whether Y RNAs target the licensing or the activating step of DNA replication. In *X. laevis*, Y RNAs function only at developmental stages that follow the midblastula transition<sup>79</sup>. Taken together, these studies suggest that ncRNAs could function in initiating DNA replication, but their mechanisms of action remain to be defined. It is possible that ncRNAs regulate DNA replication initiation by affecting G4 DNA structures. Only 20–30% of pre-RCs that are formed in G1 are activated in S phase. Moreover, pre-RCs cannot form during S phase to avoid reinitiation of DNA replication, which would lead to genomic instability. In their folded form, G4s could prevent replication initiation at pre-RCs. A short RNA transcribed by or hybridizing to the C strand of the G4 may form an R loop and favour folding of the G4 at the G strand, and DNA replication would not initiate until the G4 is unfolded. However, transcription at the G4 could also have the opposite effect and activate initiation of DNA replication. The finding that ORC can bind to G4 RNA<sup>78</sup> is compatible with both hypotheses.

### Nuclear structure and origin activation

The metazoan genome is organized into subnuclear compartments within the 3D nuclear space. The nuclear envelope, chromatin domains and replication foci are the main nuclear structures that are involved in the regulation of origin activation.

**The nuclear envelope and lamins.** The formation of the nuclear envelope around chromatin at the end of cell division is a prerequisite for origin activation and initiation of DNA replication, but not for pre-RC assembly, which can occur in the absence of nuclear membrane components<sup>80–82</sup>. This finding suggests that nuclear membrane formation is an essential step in the selection of the origins that are to be activated. Nevertheless, concentrated nuclear extracts from nuclei assembled in *X. laevis* egg extracts can initiate DNA replication without nuclear membranes<sup>83</sup>, suggesting that the main role of the nuclear membrane is to create a local concentration of replication factors. *In vitro* concentrated nuclear extracts could mimic such local concentrations at replication foci that form *in vivo* (see below).

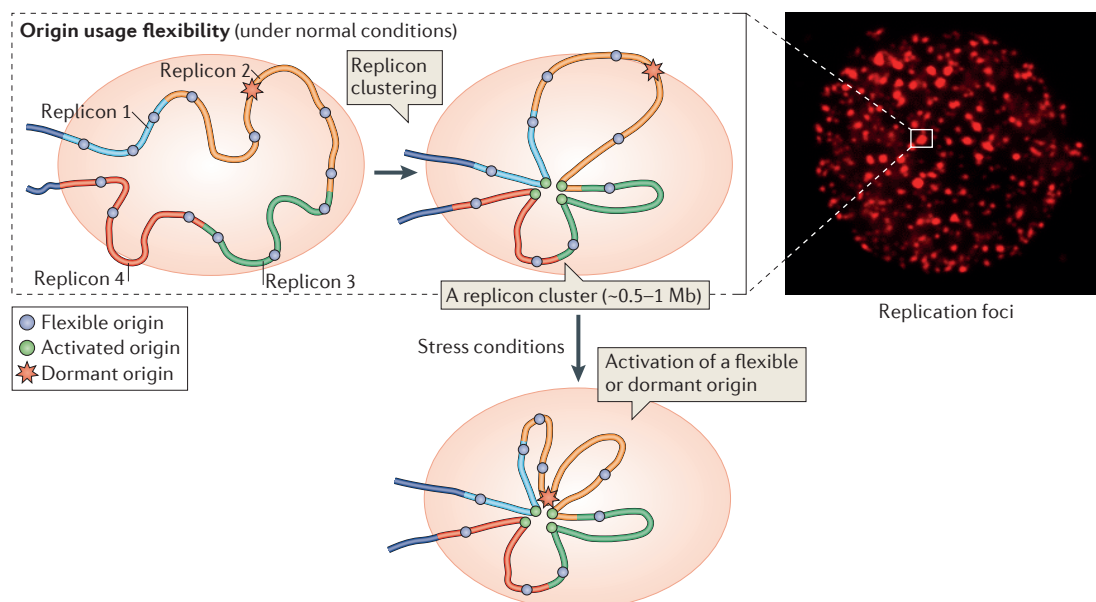
Lamins are intermediate filament proteins that form a meshwork within the internal nuclear membrane and anchor chromatin, but they are also present in the nucleoplasm. They associate with the nuclear matrix, which is still an imprecisely defined structure, and with mainly gene-poor genomic domains<sup>84</sup>. Lamins are involved in DNA replication in *X. laevis* egg extracts<sup>85</sup>. Although highly debated, the nuclear matrix and its possible role in the formation of chromatin loops could contribute to several aspects of nuclear metabolism, including DNA replication<sup>86</sup>.

**Chromosome domains.** An important feature of replication origins is that their activation is often synchronous in the several consecutive replication units that form a replication cluster (FIG. 2). The first observations by Huberman and Riggs<sup>87</sup>, together with more recent DNA combing experiments and genome-wide studies, all indicate that replication origins have different levels of organization (FIG. 2). The first level consists of the pre-RCs that are formed at all potential replication origins in a cell. The second level is the replication unit or replicon (50–120 kb in metazoan cells). Each replication unit may contain several potential pre-RCs, of which only one will be activated, thus providing flexibility in replication origin usage. This flexibility allows the use of different origins in each cell according to its specific chromatin folding<sup>88</sup> and in response to DNA damage, cell growth conditions and developmental fate. When an origin is activated in a replication unit, all the other origins from the same replication units are repressed. This occurs through a phenomenon termed negative origin interference, which was first described in yeast<sup>89</sup> and later in human cells<sup>90</sup>, and through the generation of the replication forks that move through the replicon and inactivate the other potential origins in approximately 30–60 minutes.

The third level is the association of replicons in replication clusters, which are replication domains (400 kb to 1 Mb in mammalian cells) that form replication

#### DNA combing

A method in which single DNA molecules are stretched on silanized glass. This method allows the detection of genomic abnormalities such as DNA rearrangements and is also a powerful technique for detecting the spacing between replication origins and the replication fork speed.



**Figure 2 | Organization of replication origins and origin flexibility.** The schematic shows a chromatin domain containing four consecutive replication units (shown in different colours). Each replication unit contains three to four potential flexible replication origins (blue circles) on average, as determined by genome-wide analysis coupled to DNA combing experiments and bioinformatics analysis<sup>5</sup>. These replication units interact to form a replicon cluster in which the origins that will be activated (one per replication unit; green circles) gather together within the cluster. These clusters may vary from cell to cell owing to the flexibility in choice of replication origins within the same replication unit. In a cluster, DNA replication origins that interact (green circles) fire synchronously and the cluster is identified as a replication focus in which ongoing DNA replication can be detected. Under stress conditions (for example, a replication block), flexible or dormant origins (red star) can be used to rescue DNA replication in the corresponding replicon cluster. The image on the right shows replication foci identified by biotin–dUTP pulse labelling during DNA synthesis in a nucleus. The image is reprinted with permission from REF. 192, Springer.

foci (FIG. 2). In each replicon cluster, origins fire synchronously<sup>87</sup> through a mechanism of positive interference<sup>91</sup>, in contrast to the negative interference that inhibits origins within the same replicon (FIG. 1). The replicon organization in clusters might involve chromatin looping to bring origins from different replicons into a single domain. A correlation between loop size and replicon size has been found in several species<sup>92–94</sup>. Importantly, this looping structure does not necessarily imply that an underlying structure is necessary for cluster formation. Indeed, a conglomerate of several replicons could form a structure that facilitates the simultaneous activation of the origins that it contains. The organization of the replication units in loops may also explain why only one origin per replication unit, which is located at the base of the loop, is activated (FIG. 2). Between 800 and 4,000 replication foci, each containing 4 to 6 origins on average, can be detected in a cell, depending on the microscopic resolution<sup>95–97</sup>. Clustering of different origins in single replication domains could be facilitated by cohesin, because its binding to chromatin is tightly linked to DNA replication<sup>53,98,99</sup>.

#### Cohesin

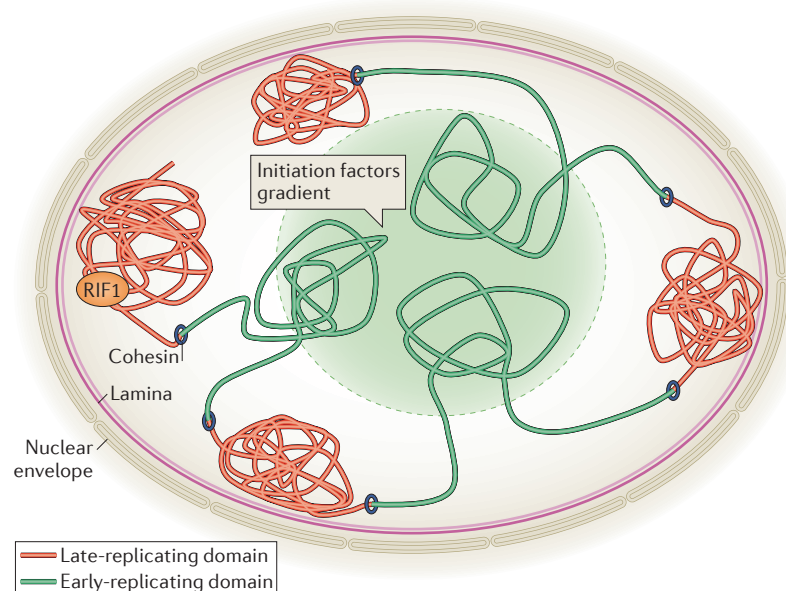
A protein complex that mediates the cohesion between the sister chromatids resulting from DNA replication and is also involved in their segregation during mitosis.

**Replication foci.** In all eukaryotes, DNA synthesis takes place at hundreds of subnuclear structures called replication foci or replication factories (reviewed in REFS 95,100). These can be identified by the incorporation of labelled deoxynucleotides as discrete focal sites

in the nucleus (FIG. 2). The biochemical nature of these foci and the mechanisms underlying their formation are poorly understood. Unlike elongation factors such as replication protein A (RPA), PCNA and DNA methyltransferase, pre-RC factors do not localize in clear replication foci, but can be detected by immunofluorescence using specific antibodies or GFP in a punctate pattern. This suggests that replication foci do not form during recognition and licensing of replication origins, but rather during or after their activation. Interestingly, the size of replication foci, obtained by PCNA staining, increases when the replication fork is blocked<sup>96</sup>, which suggests that new, flexible origins are recruited to the same foci.

Lamins are also associated with replication foci<sup>101</sup>, and their disruption inhibits the elongation phase of DNA replication but not the association of pre-RC components with chromatin<sup>102</sup>, indicating that pre-RC formation does not require specific nuclear structures. It will be important to elucidate whether it is activation of origins that allows their clustering in nuclear foci, or the inverse. Replication foci are not observed in and are not required for the replication of plasmid DNA in concentrated nuclear extracts from *X. laevis* eggs<sup>103</sup>. Conversely, they are detected when  $\lambda$ DNA is assembled in the nuclei of *X. laevis* egg extracts<sup>104</sup>. These observations suggest that the formation of the nucleus at the end of cell division introduces structural organization for DNA replication.





**Figure 3 | Nuclear organization and replication timing in mammals.** A nucleus is shown, with its lamina (pink lines) embedded in the nuclear envelope. Early replicating origins are frequently associated with highly transcribed open chromatin domains (green). By contrast, late origins are associated with silent chromatin (red) that is highly compacted and transcriptionally inactive. DNA replication starts in early-replicating domains in the interior of the nucleus. The boundaries of these domains are bound by cohesins, which are protein complexes that ensure the cohesion of sister chromatids during mitosis and meiosis and also promote inter- and intra-chromosome interactions. The timing of origin activation is regulated by several proteins, such as the telomeric protein Rap1-interacting factor (RIF1), which delays the activation of late origins by sequestering them from activation by initiator factors.

### Timing of origin activation and replication

Replication origins are not all activated at the same time, but follow a temporal programme of activation throughout the S phase, called the replication timing programme (for a review, see REF. 105). This programme is probably imposed by the complex organization and folding of chromosomes in the nucleus, the regulation of the transcriptional programmes and the concentrations of limiting replication factors. Thus, unsurprisingly, origin organization and several of the associated factors described above may influence the timing of replication.

**Early- and late-replicating domains.** The genome can be segmented into Mb-scale domains that replicate relatively synchronously at defined points during the S phase<sup>105,106</sup>. Early replication is generally observed in transcriptionally active gene-rich domains with active epigenetic marks<sup>37,107,108</sup>. These chromosomal regions are also enriched in replication origins and ORC<sup>5,53,108</sup>, possibly explaining why they replicate early in the S phase. Conversely, late replication is observed in origin-poor regions that have low gene density and are enriched in repressive epigenetic marks that are heterochromatin hallmarks. The scarcity of DNA replication initiation in these regions has been associated with common fragile sites, which are regions that tend to form secondary structures that hinder

completion of replication and are prone to breakage<sup>109</sup>. However, early-replicating fragile sites also exist, possibly owing to interference between transcription and replication<sup>110</sup>.

Late-replicating regions are located within lamin-associated-domains<sup>84,111</sup>, which indicates that nuclear structures regulate DNA replication. Importantly, the 3D organization of chromosomes is related to replication timing domains (FIG. 3). Chromosomes are segmented into structurally distinct chromatin domains in which DNA can fold so that sequences interact with other sequences within the same domain but not with sequences in adjacent domains. Interestingly, it has been observed that there is a striking overlap between these domains and replication domains<sup>112</sup>. Recently, similar Mb-scale chromatin domains that fold in a 3D space and favour internal chromatin interactions have been characterized and termed topologically associated domains (TADs)<sup>113</sup>. TADs overlap with the replication timing domains, strengthening the idea that the sets of origins within specific structural domains along a chromosome fibre are activated in a temporally regulated manner<sup>114</sup>. Early activation during the cell cycle occurs in the nuclear interior, which is more permissive to transcription, whereas late-replicating TADs reside at the nuclear periphery or in other transcriptionally repressed compartments<sup>114</sup> (FIG. 3).

**Proteins and chromatin marks affecting replication timing.** In *S. cerevisiae*, forkhead protein homologue 1 (Fkh1) and Fkh2 are global regulators of early replication<sup>51</sup>. These transcription factors bind to early-replicating origins to stimulate their activation. Moreover, Fkh1 and Fkh2 promote clustering of early origins<sup>51</sup>. This spatial reorganization, which could indicate a pre-replication focus, stimulates Cdc45 recruitment, leading to pre-IC formation for early S phase origin activation (FIG. 3).

Telomere-associated protein RIF1 is another trans-acting factor that regulates replication timing<sup>115–117</sup> (FIG. 3). RIF1 is a telomeric protein that is regulated by the DNA damage response kinases Ser protein kinase ATM (also known as ataxia telangiectasia mutated)<sup>118</sup> and Ser/Thr protein kinase ATR (also known as ataxia telangiectasia and Rad3-related protein)<sup>119</sup> that signal in response to DNA damage to activate a checkpoint that delays the cell cycle. RIF1 restricts the accessibility of mid-S phase replicating regions to CDC7, thus delaying pre-IC formation and origin activation<sup>116,117</sup>. Depletion of RIF1 results in global alterations of replication timing, with loss of the mid-S-phase replication pattern of replication foci. An early replication pattern of foci remains present during most of the S phase, followed by the late-replication foci of heterochromatin at the end of the S phase. RIF1 negatively regulates replication origin activation by preventing premature phosphorylation of MCM through directing protein phosphatase 1 to counteract DDK activity<sup>120–122</sup>. Depletion of RIF1 could, therefore, induce accumulation of phosphorylated MCM, favouring deregulated activation of replication origins.

RIF1 also regulates nuclear organization. Chromatin loops are more relaxed in the absence of RIF1 (REF. 117), which reinforces the notion of a connection between nuclear structure and replication timing. On the basis of on these data, it is expected that Hi-C maps of *RIF1*<sup>-/-</sup> cells will show substantial alterations in their chromatin structure compared to wild-type cells, reflecting the changes in replication timing that are observed.

Hp1 regulates replication timing in heterochromatin in *D. melanogaster*<sup>123</sup>, as does its orthologue Swi6 in *Schizosaccharomyces pombe*<sup>124</sup>. By promoting heterochromatin formation, Hp1 shields these chromatin regions from replication initiation and/or activation factors and thus is responsible for the very late replication of centromeric DNA in *D. melanogaster* cells<sup>123</sup>. Intriguingly, Hp1 also binds to early-replicating regions, such as *D. melanogaster* pericentromeric DNA<sup>125</sup> or the mating-type cassettes in *S. pombe*<sup>126</sup>. The early replication of these regions depends on the presence of Hp1 (REF. 123). It is believed that direct association of Hp1 with ORC<sup>71</sup> or Cdc7 (REF. 124) might stimulate licensing and activation of replication origins.

Histone acetylation is also involved in the regulation of replication timing. Constitutive heterochromatin regions (which are known as chromocentres in mouse cells) are enriched in H3K9me3 and poor in histone acetylation and replicate late. Ectopic hyperacetylation of chromocentres leads to early replication of these regions<sup>127</sup>, showing that histone acetylation has a positive role in the timing of early replication.

Replication timing can also be regulated in *cis* by nuclear factors. Telomere length regulator Taz1 is responsible for the late replication of a substantial subset of origins in fission yeast<sup>128</sup> (FIG. 3). Taz1 is the yeast orthologue of human telomeric repeat-binding factor 1 (TRF1) and TRF2 and is involved in telomere maintenance<sup>129</sup>, binding directly to the origins it regulates and suppressing loading of DNA replication regulator Sld3 and pre-IC formation. Taz1 might regulate the accessibility of late origins to Cdc7 and consequently their activation. Finally, the availability of limiting factors that are involved in activation of DNA replication also influences the timing of replication. Interestingly, several activating factors, such as Cdc45 (REFS 130, 131) and Cdc7 (REF. 132), and the CDK targets Sld2, Sld3, and Dbp11 (REF. 133) that are present in limited amounts, might become more available in the late S phase, because they are progressively recycled after early replication, increasing the probability of activation of less permissive late-replicating regions. In yeast, the 150–200 copies of rDNA, each of which contains an origin, can compete with single-copy origins for initiation<sup>134</sup>, in a reaction that is regulated by histone deacetylases<sup>135</sup>.

### Developmental control of DNA replication

About half of the human genome is replicated at different times in different cell types, highlighting the plasticity of the replication programme and a link between cell identity and timing of DNA replication<sup>111</sup>. The DNA replication timing pattern also varies among humans and is influenced by inherited genetic polymorphisms<sup>136</sup>. Moreover, within the same organism, a

substantial proportion of replication timing domains change from early to late and vice versa during cellular differentiation<sup>137</sup>.

These changes in replication timing are associated with nuclear reorganization<sup>138</sup>. Globally, changes in replication timing tend to consolidate small replication-timing domains into larger domains. It will be interesting to evaluate whether TADs that correspond to consolidated replication domains fuse together or remain independent.

Not only the timing of replication but also the usage and distribution of replication origins are regulated during differentiation and development. This has been shown in amphibians<sup>21,139</sup> and *D. melanogaster*<sup>20,140</sup> and at the immunoglobulin heavy chain locus during mouse B cell development<sup>141</sup>. Changes in origin usage seem to be linked to developmentally regulated modifications in transcriptional programmes<sup>21,139</sup>. The developmentally regulated changes in the origin pattern can also be recapitulated in experiments that mimic animal cloning in *X. laevis* eggs<sup>93</sup>. These developmental aspects of the distribution of replication origins along chromosomes might suggest an important role for replication origins in cell identity. However, it is difficult at present to say whether replication origin positioning has a direct effect on gene expression or vice versa.

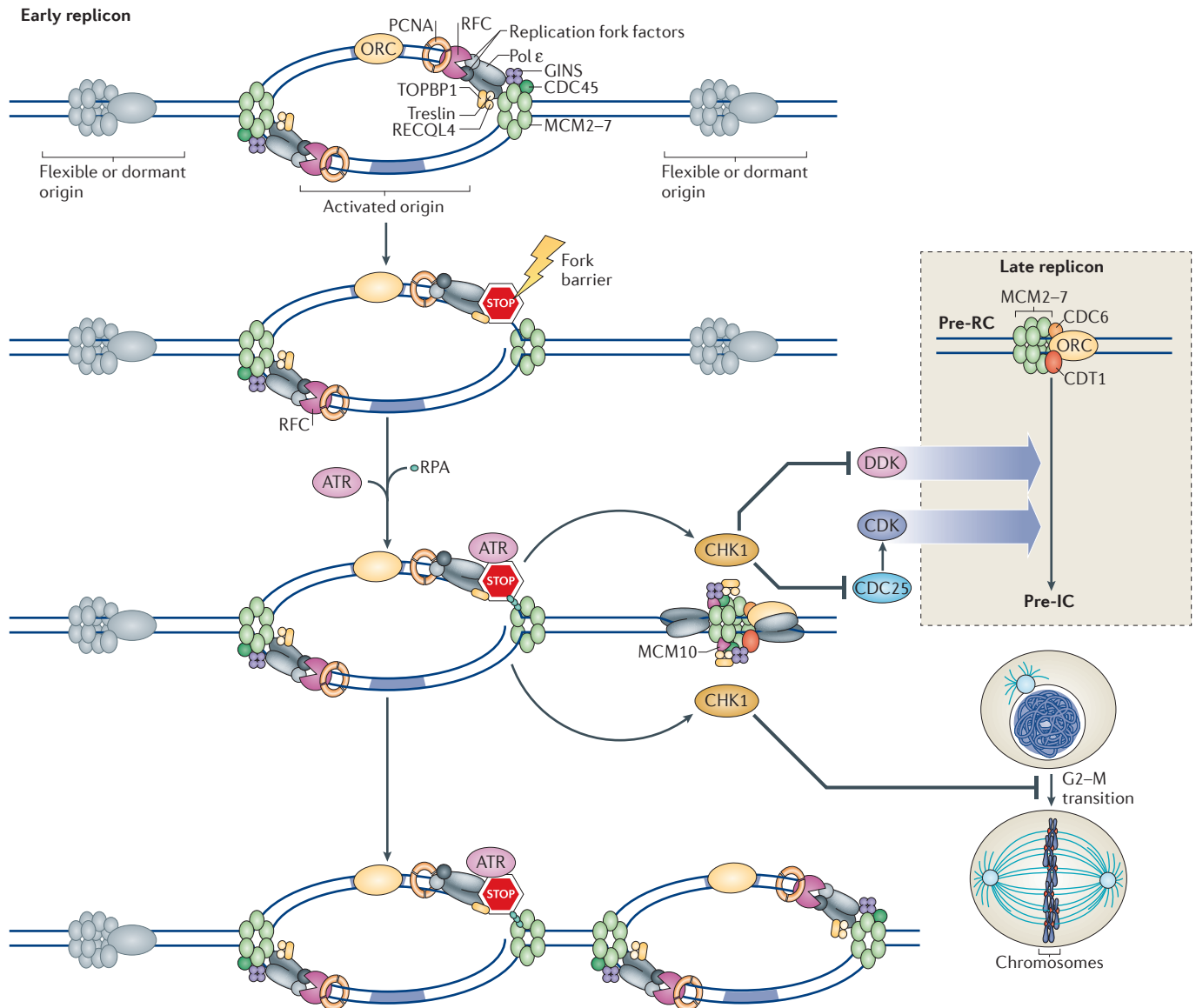
### Checkpoint control of origin activation

Different mechanisms control genomic stability, particularly by detecting abnormal DNA structures or DNA damage at replication forks and regulating cell cycle checkpoints<sup>142</sup>. In budding yeast, preventing damage-induced DNA replication fork catastrophe is a mechanism by which checkpoints preserve cell viability<sup>143</sup>. The regulation of origin activation also is a key rescue response to avoid genetic instability following replication stress. During the S phase, replication stress is sensed by the intra-S-phase checkpoint. A stalled fork induces a double and apparently paradoxical response: global inhibition of origin activation and activation of origins that are close to the stalled fork (reviewed in REFS 8, 144, 145) (FIG. 4).

**Global inhibition of origin activation.** Following replication stress, mitotic entry is delayed by checkpoint activation to facilitate DNA repair<sup>146</sup>. The molecular mechanisms of checkpoint activation are relatively well characterized. The kinases ATR and checkpoint kinase 1 (also known as Ser/Thr protein kinase CHK1) have important roles. ATR-interacting protein binds to RPA-coated single-stranded DNA and is usually described as the main sensor of this checkpoint. CHK1, which is a target of ATR, amplifies the checkpoint signal by phosphorylating and inactivating the CDC25-family phosphatases that normally activate CDKs<sup>147</sup>. This results in inhibition of CDK1 and, consequently, delay of mitotic entry<sup>148</sup> (FIG. 4). In parallel, global inhibition of origin activation is triggered by the checkpoint. Inhibition of unfired origins occurs after MCM10 loading but before recruitment of CDC45 and WD repeat and HMG-box DNA-binding protein 1 (WDHD1; the homologue of yeast Ctf4)<sup>149</sup>. Chromatin loading of CDC45, which is considered to be one of the limiting factors of origin activation,

#### Hi-C

A method to detect interactions between chromatin domains in the nucleus.



**Figure 4 | Local activation and global inhibition of origin activation following replication stress.** Following DNA damage or fork stalling, the mini-chromosome maintenance (MCM) helicase complex, which contains the six subunits MCM2–7, is uncoupled from the blocked DNA polymerases, inducing formation of stretches of single-stranded DNA that are rapidly covered by replication protein A (RPA). Then, proteins, termed the damage sensors, bind to the fork. Among them, the checkpoint kinase ATR is recruited at the stalled fork and phosphorylates checkpoint kinase 1 (CHK1), leading to the local activation of a nearby origin to complete replication of the replicon. This origin can be either a dormant origin or a flexible origin that would not normally be activated during this S phase in this cell (see BOX 2). At the same time, activation of late origins is inhibited (shaded box). It is thought that CHK1 diffuses the damage signal throughout the nucleus, by phosphorylating and inhibiting DBF4-dependent kinase (DDK) and the cyclin-dependent kinase (CDK)-activating phosphatase CDC25, thus blocking formation of the pre-initiation complex (pre-IC) (see FIG. 1). In addition, progression of the cell cycle is arrested (that is, entry into mitosis is blocked) until DNA damage is repaired and the checkpoint is switched off. CDT1, CDC10 dependent transcript 1; ORC, origin recognition complex; PCNA, proliferating cell nuclear antigen; Pol ε, DNA polymerase ε; RECQL4, ATP-dependent DNA helicase Q4; RFC, replication factor C; TOPBP1, DNA topoisomerase 2-binding protein 1.

is thus inhibited following checkpoint activation. This process involves H3K4 methylation by histone-Lys N-methyltransferase 2A (KMT2A; also known as MLL)<sup>150</sup> and the presence of treslin and MDM2-binding protein<sup>30,151</sup>. In yeast, Rad53 (which is orthologous to CHK2) phosphorylates and inhibits Sld3 and Dbf4 (REFS 152,153), which are activators required for Cdc45 loading.

**Activation of local origins during replication stress.** In contrast to the global inhibition of origins, origins that are close to stalled replication forks are activated to complete DNA replication. This ‘emergency’ pathway relies on the activation of nearby silent origins that could be either the flexible origins that were not activated in the same replication unit or dormant

origins<sup>8</sup> (FIG. 4). Replication stress caused by DNA damage or poor growing conditions<sup>154</sup> can increase the usage of flexible origins or activate a neighbouring dormant origin to complete DNA replication<sup>149</sup>.

#### *DNA damage-independent regulation by the checkpoint.*

The intra-S-phase checkpoint also regulates origin activation during the normal S phase in the absence of DNA damage and replication stress<sup>91,155</sup>. The DNA damage checkpoint programme might amplify a basal signalling network that is involved in the regulation of origin activation during the unperturbed S phase. Indeed, ATR (Mec1 in yeast) and CHK1 are essential for proper replication in metazoans and budding yeast<sup>156–161</sup>. Disruption of the genes encoding these proteins leads to disorganization of origin activation: late origins fire earlier and dormant origins become activated in the absence of stress. This over-activation of origins causes fork collapse and mitotic defects. In conclusion, the molecular mechanisms that drive coordination of origin activation seem to use the same factors that are involved in the DNA damage response (that is, the CHK1–CDC25–CDK and CHK1–CDC25–DDK pathways)<sup>162,163</sup>.

As well as the checkpoint control, other specific factors could be involved in restricting origin activity or activating dormant origins. Restriction of origin activity could be explained by a specific epigenetic signature that spreads along a cluster of origins or by a structural environment that delineates the cluster borders. In support of this explanation, the initiation sites of inactive and active origins seem to be located in the same replication factories<sup>164,165</sup>. Moreover, when dormant origins become activated after DNA damage<sup>94</sup>, the DNA loop size decreases through a mechanism similar to that observed during replication origin reprogramming in embryonic development<sup>93</sup>.

### Conclusions and perspectives

Genetic and epigenetic plasticity and usage flexibility are important characteristics of metazoan replication origins. Although origins are located at specific sites on the genome, only a subset of these potential origins is activated, and the choice of origins to be activated varies

from cell to cell. Plasticity and flexibility might ensure that replication of every DNA sequence is completed during each cell cycle, which is essential to prevent genome instability. Replication origins that were too strictly defined along chromosomes would be a hindrance to adaptation to changes in growth conditions, cell cycle contexts, replication fork blocks, developmental programmes and 3D folding of chromosomes and would probably also be a hindrance to evolution.

A third important characteristic of replication origins is their two-step activation: licensing of origins with the assembly of the pre-RC in the G1 phase, followed by their timely, organized activation during the S phase. The genetic and epigenetic signatures that are required for origin licensing and origin activation still remain unclear in metazoans, and whether some of them could be strictly specific to replication origins remains to be determined. G-rich and G4 elements are a conserved feature of metazoan replication origins, but how they regulate origin function is not known.

How replication origins are connected to chromosome structure remains another important question. Whether origins are fixed elements that structurally divide DNA in functional units, or whether they are the consequence of this segmentation, is unclear.

It also remains to be determined whether a defined set of structural proteins is necessary for assembling replication foci, or whether replication foci naturally form through the interaction of several enzymatic origin complexes. Several important proteins that are involved in origin recognition and usage have been discovered in recent years. However, particularly in metazoans, it is still unclear how these factors recognize origins and whether origin activation is regulated by a single pathway.

Unravelling the molecular details of the activation of replication origins might provide insight into the mechanisms that lead to their dysregulated activation in diseases, such as cancer, that are associated with abnormal cell proliferation. A better understanding of origin function may also pave the way for the development of improved autonomously replicating episomal vectors for gene therapy.

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**This study shows that during replication stress, the excess of the MCM complex can be used to activate DNA replication origins that are not used in a normal cell cycle.**

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## Competing interests statement

The authors declare no competing interests.